



Karm.

Karm Facility & Asset Reference Guide

Facility & Asset Reference Guide — With Basic Specs

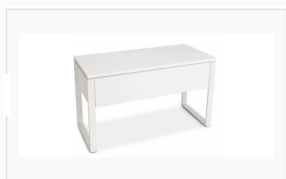
This catalogue provides a standardized reference for all approved Facilities Management equipment and consumables used across Karm locations. It serves as the official reference for procurement, replacement, maintenance, and budgeting purposes.



 Furniture & Storage	6 items
 Kitchen & Pantry Appliances	9 items
 Electrical & Power	3 items
 Climate, Bathroom & Hygiene	3 items
 HVAC Devices	1 item
 Cleaning & Hygiene Supplies	18 items
 Administrative Supplies	10 items
 Catering Supplies	16 items
 Decoration	3 items
 IT Equipment	17 items
 HSE	30 items



FM-001



Unbranded – Office Desk

Dimensions	1400 x 700 x 750 mm (W x D x H)
Material	MFC board, powder-coated steel legs
Capacity	Max load ~50 kg (worktop)
Expected Lifetime	~10 years

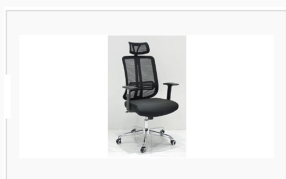
FM-002



Unbranded – Sharing / Meeting Table

Dimensions	2400 x 1000 x 750 mm (W x D x H)
Material	Laminated MDF top, wood-veneer legs
Capacity	Seats 6–8 people
Expected Lifetime	~10 years

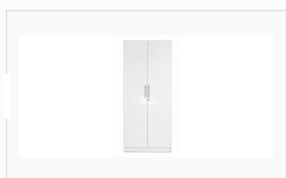
FM-003



Unbranded – Office Chair

Dimensions	650 x 650 x 1130–1220 mm (adjustable height)
Material	Mesh back, foam seat, chrome base
Capacity	Max user load ~120 kg
Expected Lifetime	~7 years

FM-004



Unbranded – Storage Cupboard

Dimensions	800 x 450 x 1800 mm (W x D x H)
Material	Melamine-faced board, hinged doors
Capacity	3 internal shelves, ~25 kg/shelf
Expected Lifetime	~12 years

FM-005



Unbranded – Drawer Cabinet

Dimensions	400 x 500 x 700 mm (W x D x H)
Material	Powder-coated steel
Capacity	5 drawers, ~10 kg/drawer
Expected Lifetime	~10 years



FM-006



Unbranded – Office Safe

Dimensions

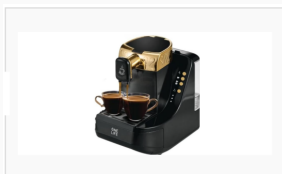
25 x 35 x 25 cm (W x D x H)

Expected Lifetime

~20 years



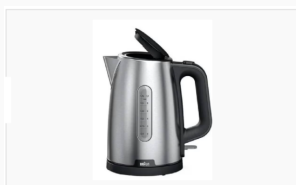
FM-007



Okka – Turkish Coffee Machine

Dimensions	300 x 250 x 350 mm (W x D x H)
Power	220–240V, ~1450W
Capacity	Water tank ~1.0 L
Expected Lifetime	~5 years
Alternative Approved	Arzum

FM-008



Braun – Electric Kettle (1.6L)

Dimensions	220 x 150 x 255 mm (W x D x H)
Power	220–240V, ~2200W
Material	Stainless steel body
Capacity	1.6 L
Expected Lifetime	~5 years
Alternative Approved	Philips

FM-009



Sharp – Microwave (23L)

Dimensions	480 x 380 x 280 mm (W x D x H)
Power	220–240V, ~1200W
Capacity	23 L cavity
Expected Lifetime	~7 years
Alternative Approved	LG

FM-010



Sharp – Refrigerator (396L)

Dimensions	595 x 650 x 1700 mm (W x D x H)
Power	220–240V, ~150W running
Capacity	396 L gross
Expected Lifetime	~10 years
Alternative Approved	LG

FM-011



Sharp – Mini Fridge (90L)

Dimensions	85 x 47 x 45 cm (W x D x H)
Capacity	90 L
Expected Lifetime	~8 years
Alternative Approved	LG



FM-012



Koldair – Floor-Standing Water Dispenser (2 Taps)

Dimensions	300 x 305 x 1010 mm (W x D x H)
Power	220–240V, ~550W (heating) / ~90W (cooling)
Capacity	Cold 3.2 L / Hot 1 L tanks
Expected Lifetime	~8 years
Alternative Approved	Fresh

FM-013



Fresh – Desktop Water Dispenser (2 Taps)

Dimensions	230 x 230 x 360 mm (W x D x H)
Power	220–240V, ~500W (heating) / ~70W (cooling)
Capacity	Cold 2.5 L / Hot 0.8 L tanks
Expected Lifetime	~6 years
Alternative Approved	Koldair

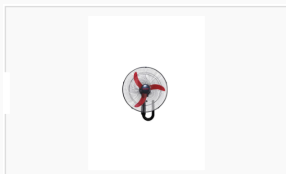
FM-014



Fresh – Stand Fan (18")

Dimensions	450 x 520 x 1540 mm (W x D x H, adjustable)
Power	220–240V, ~65W
Capacity	18" blade, 3 speeds, 90° oscillation
Expected Lifetime	~7 years
Alternative Approved	Tornado

FM-015



Fresh – Wall Mounted Fan (18")

Dimensions	500 x 450 x 550 mm (W x D x H, wall-mounted)
Power	220–240V, ~60W
Capacity	18" blade, 3 speeds
Expected Lifetime	~7 years
Alternative Approved	Tornado



FM-016



Schneider Electric – DomA41 C32

Dimensions	1-pole module, ~18 x 80 x 70 mm
Power	Rated up to 63A, 230/400V
Material	Thermoplastic housing
Expected Lifetime	~15 years
Alternative Approved	ABB

FM-017



iLock – Basic

Dimensions	~300 x 60 x 40 mm
Power	220–240V, max 13A / 3000W
Capacity	4–6 outlets
Expected Lifetime	~4 years

FM-018



iLock – Slim Triple Plug Adapter

Dimensions	~80 x 60 x 55 mm
Power	16 A, 250 V, max 3500 W
Material	Polycarbonate housing, brass pins (UK 3-pin)
Expected Lifetime	~4 years



FM-019



Duravit – Wall-Hung Toilet

Dimensions	360 x 540 x 340 mm (W x D x H)
Material	Vitreous china
Capacity	Dual flush, 3/6 L
Expected Lifetime	~20 years
Alternative Approved	Ideal Standard

FM-020



Duravit – Wash Basin

Dimensions	~400 x 400 x 150 mm (countertop bowl)
Material	Vitreous china
Expected Lifetime	~20 years
Alternative Approved	Ideal Standard

FM-021



Glade – Automatic Air Freshener

Dimensions	~120 x 80 x 100 mm
Power	Battery-operated (2 x AA)
Capacity	Refill cartridge ~269 ml
Expected Lifetime	~3 years (dispenser unit)
Alternative Approved	Air Wick



FM-022



Carrier – Split Air Conditioner

Dimensions	Indoor unit ~900 x 200 x 300 mm (W x D x H)
Power	220–240V, 1.5–2.0 Ton typical, ~1500–2200W
Capacity	Cooling 18,000–24,000 BTU/h
Expected Lifetime	~12 years
Alternative Approved	Daikin



FM-023



Glade – Automatic Spray Refill (Lavender)

Dimensions	~75 x 75 x 130 mm (canister)
Material	Aerosol canister, fits Glade auto-spray holder
Capacity	175 g / 269 ml — rated ~60 days effective freshness
Shelf Life	~24 months
Alternative Approved	Air Wick

FM-024



Areej – Hand Soap & Body Lotion Set

Dimensions	~50 x 50 x 180 mm per bottle (pump dispense)
Material	Glass/PET bottle, pump dispenser
Capacity	250 ml each (hand soap + body lotion)
Shelf Life	~24 months

FM-025



Dettol – Power Bathroom Cleaner

Dimensions	~70 x 70 x 250 mm (trigger spray bottle)
Material	PET bottle, trigger spray head
Capacity	500 ml
Shelf Life	~24 months
Alternative Approved	Clorox

FM-026



Dettol – Power Kitchen Cleaner

Dimensions	~70 x 70 x 250 mm (trigger spray bottle)
Material	PET bottle, trigger spray head
Capacity	500 ml
Shelf Life	~24 months
Alternative Approved	Clorox

FM-027



Glassy – Glass Plus Cleaner (5-in-1)

Dimensions	~90 x 90 x 280 mm (trigger spray bottle)
Material	PET bottle, trigger spray head
Capacity	Standard retail size, gift refill pack
Shelf Life	~24 months



FM-028



Mr Muscle – Toilet Cleaner

Dimensions	~90 x 90 x 300 mm (angled-neck bottle)
Material	HDPE bottle, angled pour spout
Capacity	Standard retail size (~750 ml)
Shelf Life	~24 months
Alternative Approved	Harpic

FM-029



Pledge – Multisurface Floor Cleaner (Fresh)

Dimensions	~110 x 80 x 280 mm (jug w/ handle)
Material	HDPE bottle, integrated handle
Capacity	Standard retail size (~900 ml–1 L)
Shelf Life	~24 months
Alternative Approved	Pronto

FM-030



Pledge – Furniture Polish (Original)

Dimensions	~55 x 55 x 200 mm (aerosol can)
Material	Aerosol canister
Capacity	300 ml
Shelf Life	~24 months
Alternative Approved	Pronto

FM-031



Pledge – Wooden Surfaces Cleaner (Original)

Dimensions	~110 x 80 x 280 mm (jug w/ handle)
Material	HDPE bottle, integrated handle
Capacity	Standard retail size (~900 ml–1 L), no-rinse formula
Shelf Life	~24 months
Alternative Approved	Pronto

FM-032



Raid – Flying Insect Killer (5-in-1, Odorless)

Dimensions	~55 x 55 x 220 mm (aerosol can)
Material	Aerosol canister
Capacity	Standard retail size (~300 ml)
Shelf Life	~36 months
Alternative Approved	Baygon



FM-033



Fine – Prestige Facial Tissue Box (3-Ply)

Dimensions	~250 x 130 x 70 mm (box)
Material	3-ply virgin pulp, sterilized
Capacity	Standard facial tissue box count
Shelf Life	~36 months
Alternative Approved	Softy

FM-034



Renova – Multi-Purpose Bathroom Cleaner

Dimensions	~250 x 200 x 350 mm (jerry can)
Material	formula
Capacity	20 L
Shelf Life	~24 months
Alternative Approved	Diversey

FM-035



Fema – Scouring Powder (3-in-1)

Dimensions	~70 x 70 x 160 mm (shaker bottle)
Material	deodorizes
Capacity	500 g
Shelf Life	~24 months

FM-036



Pro Wash – All Purpose Cleaner

Dimensions	~80 x 80 x 280 mm (bottle)
Material	PET bottle, multi-use surface/dish cleaner
Capacity	1050 g (1000 g + 50 g free)
Shelf Life	~24 months

FM-037



General – Lavender Multi-Surface Cleaner

Dimensions	~110 x 80 x 280 mm (jug w/ handle)
Material	formula
Capacity	Standard retail size (~800 ml–1 L)
Shelf Life	~24 months



FM-038



Cleanway – Leather Polish (Spray)

Dimensions	~60 x 60 x 200 mm (trigger spray bottle)
Material	care
Capacity	Standard retail size
Shelf Life	~24 months

FM-039



Speed – Drain Opener (Destroys Lumps & Odors)

Dimensions	~60 x 60 x 170 mm (angled-neck bottle)
Material	drain opener
Capacity	300 g
Shelf Life	~24 months

FM-040



Fine – Comfort Silky Soft Toilet Paper

Dimensions	Standard roll
Material	2-ply virgin pulp tissue
Capacity	5 + 1 free rolls per pack
Shelf Life	~36 months
Alternative Approved	Softy



FM-041



Energizer – AA Rechargeable NiMH Batteries

Dimensions	AA size, 14.5 x 50.5 mm per cell
Material	Nickel-Metal Hydride (NiMH), rechargeable
Capacity	1.2V per cell, 2300 mAh — pack of 4
Shelf Life	Holds charge ~12 months; up to 1000 recharges
Alternative Approved	Panasonic Eneloop

FM-042



ROTO – Liquid Ball Pen (Blue)

Dimensions	~140 x 10 mm (length x barrel)
Material	Plastic barrel, tungsten-carbide ball tip
Capacity	0.7 mm tip, blue liquid ink - pack of 3
Shelf Life	~24 months

FM-043



Kangaro – Staples (24/6)

Dimensions	24/6 (6 mm leg length)
Material	Galvanized steel
Capacity	~1,000 staples/box; binds 2-30 sheets
Expected Lifetime	Indefinite (shelf-stable metal)

FM-044



Multi Office – A4 Copy Paper (80 g/m2)

Dimensions	210 x 297 mm (A4)
Material	80 g/m2 multipurpose white paper, ECF
Capacity	Ream of 500 sheets
Shelf Life	~24 months
Alternative Approved	Double A

FM-045

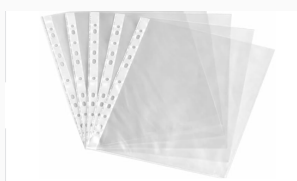


Multi Office – A3 Copy Paper (80 g/m2)

Dimensions	297 x 420 mm (A3)
Material	80 g/m2 multipurpose white paper
Capacity	Box of 5 reams x 500 sheets
Shelf Life	~24 months
Alternative Approved	Double A



FM-046



Unbranded – Transparent Punched Pockets (A4)

Dimensions	310 x 235 mm (A4, 11-hole)
Material	Clear polypropylene, glide-clear finish
Capacity	Pack of 5; multi-punched for ring binders
Shelf Life	Indefinite (inert plastic)

FM-047



Kores – K-Marker XW1 Whiteboard Marker

Dimensions	~140 x 16 mm per marker
Material	Plastic barrel, round tip, dry-erase ink
Capacity	Set of 4 colours (black, blue, red, green)
Shelf Life	~24 months
Alternative Approved	Pilot

FM-048



Unbranded – Magnetic Whiteboard Eraser

Dimensions	~110 x 55 x 30 mm
Material	ABS shell, felt wiping pad, magnetic back
Expected Lifetime	~24 months

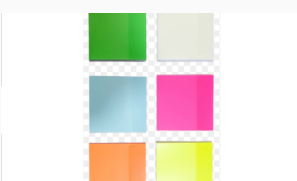
FM-049



Unbranded – Sticky Arrow Page Markers

Dimensions	~45 x 12 mm per arrow flag
Material	Self-adhesive film, neon colours
Capacity	5 colours, ~25 sheets each
Shelf Life	~12 months

FM-050



Unbranded – Square Sticky Notes (Neon)

Dimensions	~76 x 76 mm
Material	Self-adhesive paper, assorted neon colours
Capacity	~100 sheets per pad
Shelf Life	~12 months



FM-051



Puvana – Bottled Drinking Water

Dimensions	~60 x 60 x 200 mm (bottle)
Material	PET bottle
Capacity	Standard retail size (~500 ml)
Shelf Life	~12 months

FM-052



iSiS – Ginger & Cinnamon Tea

Dimensions	~120 x 60 x 130 mm (box)
Material	Organic herbal tea bags, pesticide-free
Capacity	20 tea bags
Shelf Life	~18 months

FM-053



iSiS – Green Tea

Dimensions	~120 x 60 x 130 mm (box)
Material	Organic green tea bags
Capacity	12 tea bags
Shelf Life	~18 months

FM-054



El Arosa – Black Tea

Dimensions	~140 x 65 x 90 mm (box)
Material	Premium black tea bags
Capacity	25 tea bags
Shelf Life	~18 months

FM-055



iSiS – Cinnamon Tea

Dimensions	~120 x 60 x 130 mm (box)
Material	100% natural cinnamon tea bags
Capacity	12 tea bags
Shelf Life	~18 months



FM-056



iSiS – Hibiscus Tea

Dimensions	~120 x 60 x 130 mm (box)
Material	100% natural hibiscus tea bags
Capacity	20 tea bags
Shelf Life	~18 months

FM-057



iSiS – Peppermint Tea

Dimensions	~120 x 60 x 130 mm (box)
Material	100% natural peppermint tea bags
Capacity	20 tea bags
Shelf Life	~18 months

FM-058



iSiS – Anise Tea

Dimensions	~120 x 60 x 130 mm (box)
Material	100% natural anise tea bags
Capacity	20 tea bags
Shelf Life	~18 months

FM-059



Nescafe Classic – Instant Coffee

Dimensions	~95 x 95 x 130 mm (glass jar)
Material	Soluble instant coffee, rich Arabica
Capacity	190 g
Shelf Life	~24 months

FM-060



Shaheen Coffee – Turkish (Plain, Light Roast)

Dimensions	~150 x 100 x 220 mm (stand-up pouch)
Material	Arabica/Robusta blend, finely ground
Capacity	Resealable pouch, light roast
Shelf Life	~12 months



FM-061



**Shaheen Coffee – Turkish
(Cardamom, Dark Roast)**

Dimensions	~150 x 100 x 220 mm (pack)
Material	100% Arabica with cardamom, finely ground
Capacity	Resealable pack, dark roast
Shelf Life	~12 months

FM-062



**Shaheen Coffee – Turkish
(Cardamom, Medium Roast)**

Dimensions	~150 x 100 x 220 mm (pack)
Material	Arabica/Robusta with cardamom, finely ground
Capacity	Resealable pack, medium roast
Shelf Life	~12 months

FM-063



Almarai – Long-Life Skimmed Milk

Dimensions	~90 x 90 x 300 mm (bottle)
Material	UHT skimmed milk + vitamins, PET bottle
Capacity	1.5 L
Shelf Life	~6 months (UHT, unopened)

FM-064



Almarai – Fresh Milk

Dimensions	~85 x 85 x 250 mm (bottle)
Material	Pasteurized cow milk, PET bottle
Capacity	~1 L
Shelf Life	~7 days (refrigerated)

FM-065



Imtenan – Clover Honey (Natural)

Dimensions	~95 x 95 x 150 mm (glass jar)
Material	100% natural clover-blossom honey
Capacity	800 g
Shelf Life	~24 months



FM-066

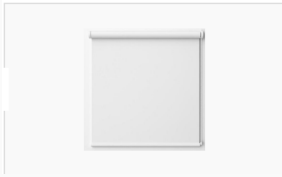


**Puvana – Water Dispenser Bottle
(Large)**

Dimensions	~300 x 300 x 600 mm (approx)
Material	PET disposable dispenser bottle
Capacity	Large cooler bottle (~12-19 L)
Shelf Life	~12 months



FM-067

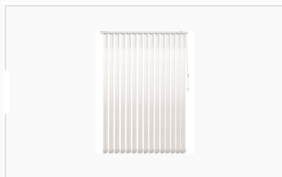


Unbranded – Rolling Curtain

Dimensions 300 x 350

Expected Lifetime ~8 years

FM-068

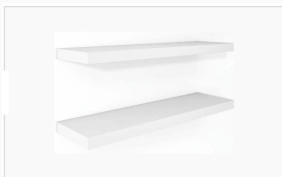


Unbranded – Strip Curtain

Dimensions 300 x 350

Expected Lifetime ~5 years

FM-069



Unbranded – Wall Shelving

Dimensions 200 x 40 x 50

Capacity 2 shelves

Expected Lifetime ~10 years



FM-070



**Lenovo – ThinkPad E14 Gen 6
(Laptop)**

Dimensions	~324 x 223 x 18 mm (W x D x H)
Power	65W USB-C adapter, 100–240V
Capacity	16 GB RAM, 512 GB SSD
Expected Lifetime	~4 years
Alternative Approved	Dell Latitude / HP ProBook

FM-071



Dell – P2422H / S2421HN (Monitor)

Dimensions	~539 x 210 x 480 mm (W x D x H, with stand)
Power	100–240V, ~21W typical
Capacity	23.8–24", 1920 x 1080 Full HD
Expected Lifetime	~5 years
Alternative Approved	HP / ViewSonic

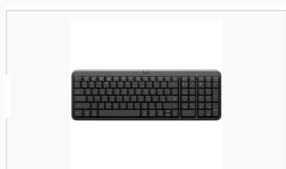
FM-072



HP – LaserJet MFP M130fw (Printer)

Dimensions	~410 x 380 x 320 mm (W x D x H)
Power	220–240V, ~200W average
Capacity	Up to 23 ppm, 35-sheet ADF
Expected Lifetime	~5 years
Alternative Approved	Ricoh MP C3002 / Canon PIXMA G3411

FM-073



**Logitech / A4Tech – Keyboard &
Mouse**

Dimensions	~442 x 142 mm (keyboard)
Power	2 x AA/AAA (wireless) or USB (wired)
Capacity	Full-size keyboard + optical mouse ~1000 DPI
Expected Lifetime	~5 yrs (keyboard) / ~3 yrs (mouse)
Alternative Approved	HP / Dell wired sets

FM-074

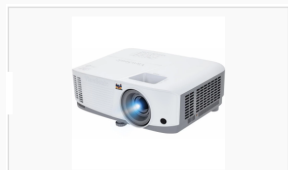


HP – fi-7160 (Document Scanner)

Dimensions	~297 x 172 x 181 mm (W x D x H)
Power	100–240V, ~35W operating
Capacity	Up to 60 ppm / 120 ipm duplex, 80-sheet ADF
Expected Lifetime	~5 years
Alternative Approved	Canon / Epson sheet-fed scanner



FM-075

**ViewSonic – PA503X (Projector)**

Dimensions	~302 x 224 x 97 mm (W x D x H)
Power	100–240V, ~245W operating
Capacity	3,800 ANSI lumens, XGA 1024x768
Expected Lifetime	~15,000 hrs lamp (Eco mode)
Alternative Approved	Epson / BenQ projector

FM-076

**Cisco – Catalyst 3750X/3850X (Switch)**

Dimensions	~445 x 292 x 44 mm (1U rack, W x D x H)
Power	100–240V, ~350W typical
Capacity	48x Gigabit ports + SFP uplinks
Expected Lifetime	~7 years
Alternative Approved	HPE Aruba / Juniper switch

FM-077

**Fortinet – FortiGate 100E (Firewall)**

Dimensions	~432 x 300 x 44 mm (1U rack, W x D x H)
Power	100–240V, ~60W typical
Capacity	7.4 Gbps firewall throughput, 2M sessions
Expected Lifetime	~5–7 years
Alternative Approved	Sophos / Palo Alto firewall

FM-078

**TP-Link Omada – OC300 + EAP245 (Access Point)**

Dimensions	~200 x 200 x 43 mm (ceiling mount)
Power	802.3af PoE, ~12.9W max
Capacity	AC1750 dual-band Wi-Fi
Expected Lifetime	~5 years
Alternative Approved	Ubiquiti UniFi AP

FM-079

**Hikvision – DS-7608NI-K2 (NVR)**

Dimensions	~315 x 242 x 48 mm (1U, W x D x H)
Power	100–240V, ~25W (excl. HDD)
Capacity	8-ch, 2-bay HDD up to 16 TB each
Expected Lifetime	~5 years
Alternative Approved	Dahua NVR



FM-080



Hikvision – DS-2CD1123G0E-I (IP Camera)

Dimensions	~114 mm dia. x 87 mm (H)
Power	12VDC or PoE 802.3af
Capacity	2 MP, 1920x1080, up to 30 m IR
Expected Lifetime	~5 years
Alternative Approved	Dahua IP camera

FM-081



Dell PowerEdge – R730xd (Server)

Dimensions	~482 x 714 x 87 mm (2U rack, W x D x H)
Power	Dual 750–1100W PSU, 100–240V
Capacity	Dual Xeon E5-2600, up to 24x SFF bays
Expected Lifetime	~7 years
Alternative Approved	HPE ProLiant server

FM-082



Cisco – C1111-8P (Router)

Dimensions	~305 x 216 x 44 mm (1U, W x D x H)
Power	100–240V, ~30W typical
Capacity	8x GE ports, dual WAN, VPN/SD-WAN
Expected Lifetime	~7 years
Alternative Approved	MikroTik / Juniper router

FM-083



APC – SRV10KI Smart-UPS (UPS)

Dimensions	~250 x 705 x 770 mm (tower, W x D x H)
Power	10 kVA / 8 kW, online double-conversion
Capacity	Runtime depends on external battery pack
Expected Lifetime	~8–10 yrs (battery ~5 yrs)
Alternative Approved	Eaton / Schneider UPS

FM-084



ZKTeco – MB2000/MB20 (Attendance)

Dimensions	~175 x 145 x 45 mm (W x D x H)
Power	12VDC, ~500 mA
Capacity	Thousands of fingerprint / face templates
Expected Lifetime	~5 years
Alternative Approved	Hikvision / Anviz terminal



FM-085

**LG – 55UT7550 (4K UHD Smart TV)**

Dimensions	~1230 x 710 x 60 mm (W x H x D, without stand)
Power	100–240V, ~120W typical / <0.5W standby
Capacity	55", 3840 x 2160 4K UHD, webOS Smart TV
Expected Lifetime	~7 years
Alternative Approved	Samsung / Hisense

FM-116

**Klearlook – Foldable Laptop Stand (11–17.3")**

Dimensions	~260 x 225 x 45 mm (W x D x H, folded)
Material	Aluminum alloy frame, non-slip silicone pads
Capacity	Fits 11–17.3" laptops/tablets; 6-position height adjustment
Expected Lifetime	~5 years
Alternative Approved	Nulaxy / Lamicall laptop stand



FM-086



Fire Extinguisher

Category	Fire Safety
Brand	Approved brands per vendor scope: BAVARIA, METORY, FREDEX (portable CO ₂ / ABC dry powder extinguishers); SIEX, LPG, TYCO (45 kg standard CO ₂ cylinder systems)
Model / Type	Portable CO ₂ gas extinguisher — 2 kg (general use / inverters), 6 kg, 10 kg; Portable ABC Dry Powder extinguisher — 3 kg, 6 kg; Standard 45 kg CO ₂ cylinder for room-level suppression (MDB rooms, store caravans) with weighing scale, Portable Foam 50L,90L
Specification / Standard	Complies with NFPA standards and the Egyptian fire code; each unit fitted with thermal tube, pressure gauge and all accessories; inspected against KS-HSE-PR-EMR-01-FM-02 checklist (location, visibility, seal, charge, corrosion, nozzle, production year)

FM-087



Smoke Detector

Category	Fire Safety
Model / Type	Optical smoke detector — component of the room-level CO ₂ /FM200 fire detection, alarm & suppression system (MDB/transformer/control rooms); standalone smoke fire detector for individual areas (e.g. store caravans)
Specification / Standard	Standalone units specified as "UL listed"; system-integrated units form part of a two-zone, double-knock fire detection & alarm control panel; overall system to comply with NFPA standards

FM-088



Emergency Exit Light

Category	Fire Safety
Specification / Standard	EN 1838 (Emergency Lighting) / ISO 30061, or NFPA 101 (Life Safety Code) egress-lighting requirements — not itemized in any KarmSolar procedure.

FM-089



First Aid Kit

Category	Emergency / Medical
Model / Type	Standard site First Aid box containing: gauze pads (4x4 in), two large gauze pads (8x10 in), box of adhesive bandages, gauze roller bandage (2 in), triangular bandages, wound-cleaning towelettes, scissors, tweezers, adhesive tape, latex gloves, resuscitation equipment (bag/airway/pocket mask), elastic wraps, splint, fire blanket, eye wash (normal saline), Dermazin (burn) cream, dryness solution, Arabic emergency-assistance instruction booklet, and the First Aid box/bag itself
Specification / Standard	Per KS-HSE-PR-EMR-01-FM-03 (First Aid Kit form, Issue 05, 01-Sep-2025); First Aiders must keep kits fully equipped, accessible and regularly inspected for availability/expiry (Sec 4.4.4 of KS-HSE-PR-EMR-01)



FM-090



Fire Alarm Control Panel

Category	Fire Safety
Model / Type	Fire detection, alarm & extinguishing control panel — two-zone, double-knock configuration; includes manual release station, abort switch, manual call points/break-glass units, and alarm bell with flashing light
Specification / Standard	Complies with NFPA standards and the Egyptian fire code; site-specific Fire Alarm System design required per as-built architectural drawings (per BOQ scope for Switch Gear Marsa Alam & Karm Water Station)

FM-091



Safety Signage (Wet Floor / Warning)

Category	Signage
Specification / Standard	ISO 7010 (Graphical symbols — Safety colours and safety signs) or the applicable Egyptian OSH signage regulation.

FM-092



Safety Shoes

Category	PPE
Model / Type	S3 — steel/composite toe cap, penetration-resistant midsole, water-resistant upper, energy-absorbing heel [recommended class]
Specification / Standard	EN ISO 20345 ("or equivalent" per Sec 3.3); 1-year replacement interval; mandatory minimum PPE on all construction sites (Sec 3.2)

FM-093



Vests

Category	PPE
Model / Type	Class 2 (minimum); Class 3 recommended for work near live traffic/moving plant [recommended class]
Specification / Standard	EN ISO 20471:2013; 6-month replacement interval; required per Sec 3.9 wherever there is risk near roadways/moving traffic or plant

FM-094



Safety Harness

Category	PPE
Brand	3M & Camb or Equivalent
Model / Type	Full-body harness for fall arrest / work positioning
Specification / Standard	EN 361 (fall arrest), EN 358 (work positioning), EN 813 (sit harness); 5-year replacement interval; mandatory above 1.8 m fall exposure (Sec 3.8); subject to periodic third-party inspection (Sec 3.8.2)



FM-095



Mechanical Gloves (Cut / Abrasion Resistant)

Category	Hand Protection
Model / Type	EN 388 rating 4121 (Abrasion 4 / Cut 1 / Tear 2 / Puncture 1) — typical general-purpose level [recommended rating]
Specification / Standard	EN 420 / EN 388; 1-month replacement interval; for heavy machinery, elevated work, rigging, mechanical work, concrete work, manual handling

FM-096



Chemical-Resistant Gloves

Category	Hand Protection
Model / Type	EN 374 Type B (or per chemical breakthrough time) — matched to the specific chemical per MSDS
Specification / Standard	EN 420 / EN 374; 6-month replacement interval; for painting and works involving chemicals (Sec 3.5)

FM-097



Dielectric Gloves — Class 00 (Low Voltage)

Category	Hand Protection
Model / Type	Class 00, max use voltage 1,000V AC (ELV work)
Specification / Standard	EN/IEC 60903 Class 00; 3-year replacement interval; for low-voltage electrical works

FM-098



Dielectric Gloves — Class 2 (Medium Voltage)

Category	Hand Protection
Model / Type	Class 2, max use voltage 17,000V AC (proof-tested to 20,000V)
Specification / Standard	EN/IEC 60903 Class 2; requires periodic electrical proof-test/re-certification in addition to visual inspection

FM-099



Safety Helmets

Category	Head Protection
Brand	3M & MSA & Or equivalent
Model / Type	Type II hard hat, EN 397 with 440V a.c. electrical insulation option [recommended option — not itemized in KS-HSE-PR-PPE-01]
Specification / Standard	ANSI Z89.1 Class E (Electrical) and EN 397; 5-year replacement interval; mandatory minimum PPE on all construction sites (Sec 3.2)



FM-100



Foundry Footwear

Category	Foot Protection
Model / Type	S3 + HRO (heat/hot-contact resistant outsole) [typical class for hot-exposure footwear — not itemized in KS-HSE-PR-PPE-01]
Specification / Standard	EN ISO 20345; 3-year replacement interval; for concrete work, post-construction basic cleaning, works involving chemicals

FM-101



Safety Lanyard / Rope (Shock Absorbing)

Category	Fall Protection
Brand	3M & Camb or Equivalent
Model / Type	Energy-absorbing lanyard rated to support 5,400 lb (2,450 kg)
Specification / Standard	EN 354 (lanyards) / EN 355 (energy/shock absorbers); 5-year replacement interval; for working at elevated locations and rigging

FM-102



Rope Grab / Fall Arrestor

Category	Fall Protection
Brand	3M & Camb or Equivalent
Model / Type	Fall arrestor for rigid or flexible anchorage line/rail
Specification / Standard	EN 353-1 (rigid anchorage line/rail) / EN 353-2 (flexible anchorage line/rope); subject to periodic third-party inspection (Sec 3.8.2)

FM-103



Fall Arrest Equipment Accessories (Connectors, Arrestors, Anchors)

Category	Fall Protection
Brand	3M & Camb or Equivalent
Model / Type	Retractable line fall arrestors, karabiner connectors, complete fall arrest systems, rigid/flexible-line fall arrestors, and anchors (Types A/B/C/D)
Specification / Standard	EN 360 (retractable arrestors) / EN 362 (connectors) / EN 363 (complete systems) / EN 353-1 & 353-2 (line arrestors) / EN 795 (anchors)

FM-104



Work Positioning Equipment (Descenders, Rope Clamps, Pulleys, Ropes)

Category	Fall Protection
Brand	3M & Camb or Equivalent
Model / Type	Sit harnesses, work-positioning lanyards (pole straps), descenders, rope clamps (ascenders), pulleys, industrial rope-access/climbing ropes
Specification / Standard	EN 358 (sit harnesses / lanyards) / EN 341 (descenders) / EN 567 (rope clamps) / EN 12278 (pulleys) / EN 1891 (ropes)



FM-105



**Other Working-at-Height PPE
(Mountaineering Helmets, Slings,
Quick Links)**

Category	Fall Protection
Brand	3M & Camb or Equivalent
Model / Type	Mountaineering helmets, maillons (quick links), rigging/man-made-fibre slings, wire slings
Specification / Standard	EN 397 (construction helmets, already covered under Safety Helmets) / EN 12492 (mountaineering helmets) / EN 12275 (maillons) / EN 566, EN 795B, EN 1492-2 (rigging slings) / BS 1290 (wire slings)

FM-106



Earplugs

Category	Hearing Protection
Model / Type	Disposable or reusable earplugs, NRR suited to noise level
Specification / Standard	EN 352-2 / EN 458; 1-month replacement interval; mandatory wherever continuous noise levels are 85 dBA or greater (Sec 3.7)

FM-107



Earmuffs

Category	Hearing Protection
Model / Type	Ear-defender earmuffs (standalone or helmet-attached)
Specification / Standard	EN 352-1 / EN 352-3 / EN 458; 6-month (normal) or 3-month (humid/extreme climate) replacement interval

FM-108



**Safety Glasses (Transparent /
Impact-Resistant)**

Category	Eye & Face Protection
Model / Type	EN 166 Optical Class 1, Impact Rating F (low-energy impact), with rigid side shields [class recommended]
Specification / Standard	EN 166 / ANSI Z87.1; 2-year replacement interval; for rigging, painting, cleaning, chemical handling and other assessed tasks

FM-109



**Chemical / Splash-Resistant
Goggles**

Category	Eye & Face Protection
Model / Type	Sealed, indirect-vent goggles, EN 166 marking '3' (liquid droplet/splash) [marking recommended]
Specification / Standard	ANSI Z87.1; 2-year replacement interval; for handling irritants, corrosives, caustics, acids, toxics, molten metal or tar (Sec 3.4.2)



FM-110



Face Shield

Category	Eye & Face Protection
Model / Type	Worn together with primary eye protection underneath (not a standalone substitute)
Specification / Standard	ANSI Z87.1; 2-year replacement interval; for chipping, scraping, blowing, pumping concrete, power sawing, grinding, pneumatic tool use (Sec 3.4.3)

FM-111



Welding Hood

Category	Eye & Face Protection
Model / Type	Filtered lens, Shade No. 10 or darker; "flip-up" window with secondary clear lens recommended
Specification / Standard	ANSI Z87.1; 2-year replacement interval; worn together with a hard hat for hot slag/overhead protection (Sec 3.4.4)

FM-112



Hygiene / Dust Mask

Category	Respiratory Protection
Model / Type	Disposable filtering half mask — FFP1 (nuisance dust) or FFP2 (finer particulate/fume) per task
Specification / Standard	EN 149; single use; for hot works/welding, painting, cleaning, chemical work, adverse weather (dust/wind) (Sec 3.10)

FM-113



Respirator

Category	Respiratory Protection
Model / Type	Tight-fitting respirator, FFP1-rated cartridge/filter (as stated in KS-HSE-PR-PPE-O1 — confirm adequacy per hazard)
Specification / Standard	EN 149 FFP1; 3-year unit life; fit test and pulmonary-function test mandatory; wearer must be clean-shaven in seal area (Sec 3.10)

FM-114



Welding Leather Gloves

Category	Welding Protection
Model / Type	Heat-resistant leather welding gauntlets
Specification / Standard	EN 388 / EN 407 / EN 12477; 1-year replacement interval; for hot works/welding



FM-115

**Welding Leather Apron**

Category	Welding Protection
Model / Type	Leather apron, Class 1
Specification / Standard	ISO 11611:2007 Class 1; 1-year replacement interval; protection from heat transfer, spatter and arc welding exposure